How MIDI Maze Initializes Game Play Over the Wire

This document describes the MIDI Maze game setup communication process as implemented in the original MIDI Maze ROM. It covers only the pre-game initialization and configuration exchange that occurs over the MIDI serial link. Gameplay data exchange is explicitly excluded.

1. Network Model Overview

MIDI Maze uses a unidirectional serial ring network operating at 31,250 baud (MIDI rate). Each Atari forwards every received byte to the next machine, forming a logical token ring. All communication consists of raw bytes; there is no packet framing beyond command bytes and strict ordering expectations.

One machine becomes the Master. All others act as Slaves. The Master is responsible for initializing the ring, distributing game configuration, and issuing the command that begins gameplay.

2. Timing and Error Behavior

During setup, MIDI Maze enforces a short receive deadline of 15 system ticks using the OS real-time clock (RTCLOK). This corresponds to approximately 250 ms on NTSC systems (60 Hz) or 300 ms on PAL systems (50 Hz). If expected data does not arrive before the deadline, the setup process aborts with an I/O timeout error.

Additional setup errors may be triggered by unexpected command bytes, missing markers, or inconsistent ring counts. Errors are detected immediately and halt the setup process.

3. Command Summary

Command Name	Hex Value	Purpose
CMD_INIT_RING	$80	Initialize ring and assign station IDs
CMD_SETUP_BOTS	$84	Begin game configuration broadcast
MARKER_SETUP_PAYLOAD	$83	Required marker preceding setup payload
CMD_START_GAME	$87	Transition from setup to gameplay

4. Setup Sequence (Step-by-Step)

Step 1: Ring Initialization (CMD_INIT_RING)
The Master sends CMD_INIT_RING ($80), followed by a station counter byte ($01). Each Slave assigns itself a station ID from this value, increments it, and forwards it. When the counter returns, the Master determines the number of participating machines.

Step 2: Game Configuration Broadcast (CMD_SETUP_BOTS)
The Master sends CMD_SETUP_BOTS ($84), which places all Slaves into setup-receive mode. This is immediately followed by a required marker byte ($83) and a fixed-size configuration payload.

Step 3: Configuration Payload
The payload defines shared random seed state and the number of computer-controlled opponents of each type. Each Slave stores the values locally and forwards them unchanged.

5. Configuration Payload Format

Byte Index	Name	Description
0	PRNG_Seed_High	High byte of shared random number generator seed
1	PRNG_Seed_Low	Low byte of shared random number generator seed
2	BotCount_Target	Number of Target-type computer players
3	BotCount_Drone	Number of Drone-type computer players
4	BotCounts_Packed	High nibble: Nasty count, Low nibble: Ninja count

If the MARKER_SETUP_PAYLOAD byte ($83) is not received immediately before this payload, the setup process fails with an error.

6. Transition to Gameplay

After successful configuration exchange and internal validation, the Master issues CMD_START_GAME ($87). All Slaves receiving this command exit setup mode and enter gameplay initialization.

No gameplay data is transmitted before this command.

7. Variable Name Reference

Human-Readable Name	Disassembly Address
playerCountHuman	L396B
PRNG_Seed_Low	L3969
PRNG_Seed_High	L396A
BotCount_Target	L3EED
BotCount_Drone	L3EEE
BotCount_Ninja	L3EEF (low nibble)
BotCount_Nasty	L3F13 (high nibble)